

SAMRUDDHI HEALTHCARE INTELLIGENCE PLATFORM

Unified UML Architecture & Systems Engineering Specification

Project: Samruddhi Healthcare Intelligence	Document: System Architecture Dossier
Domain: Clinical Systems & Predictive AI	Standard: OMG UML 2.5 Specification
Hosting Target: Render Cloud • MongoDB / PostgreSQL	Structure: Easy-to-Hard (Functional → Dynamic → Static → Physical)

1. Executive Summary & Project Introduction

The **Samruddhi Healthcare Intelligence Platform (Samruddhi Healthcare Intell)** is a state-of-the-art clinical management and epidemiological forecasting platform designed for modern healthcare institutions. The system synergistically combines core hospital operations—such as multi-department patient queuing, live bed allocation, doctor roster management, and digital prescription issuance—with an intelligent AI-driven disease surge prediction engine.

By fusing live environmental metrics (Air Quality Index - AQI, Ambient Temperature, Humidity) with inpatient clinical diagnoses, the platform anticipates localized outbreaks (e.g., Dengue, Malaria, respiratory ailments) before they overwhelm emergency resources. Furthermore, the architecture enforces a zero-trust patient privacy paradigm by securing Electronic Health Record (EHR) access through real-time, time-bound SMS OTP verification.

2. Key Architectural Pillars

- **Zero-Trust EHR Access Control:** Doctors cannot view sensitive patient history without an explicit, time-limited OTP authorized directly by the patient.
- **Predictive AI & Environmental Telemetry:** Autonomous ingestion of weather and air quality feeds to correlate atmospheric conditions with disease trends.
- **Dynamic Resource & Bed Allocation:** Real-time tracking of general, ICU, and emergency bed inventory with instant allocation capabilities.
- **Automated Advisory Engine:** Generates proactive clinical advisories for hospital administrators to scale medical staffing and pharmaceutical stockpiles ahead of seasonal spikes.

3. UML Diagram Progression & Engineering Flow

This dossier follows a structured software engineering pedagogical flow, progressing from high-level operational concepts to low-level structural and physical topology:

- **1. Use Case Diagram (Functional View):** Scope of human actors, external services, and core system boundaries.
- **2. Activity Diagram (Operational Workflow):** Step-by-step consultation lifecycle, OTP validation branch, and asynchronous AI correlation hook.
- **3. Sequence Diagram (Dynamic Interaction):** Chronological message exchange, OTP transmission, and conditional EHR retrieval.
- **4. Class Diagram (Static Domain Model):** Domain entities, attributes, methods, aggregations, compositions, and multiplicity constraints.
- **5. Deployment Diagram (Physical Infrastructure):** Physical cloud hosting topology on Render, microservices, databases, and network security protocols.

Diagram 1: System Use Case Model (Functional Scope)

The Use Case diagram establishes the behavioral boundary of the **Samruddhi Healthcare Intelligence Platform**, mapping primary human actors and external cloud microservices to system functionalities.

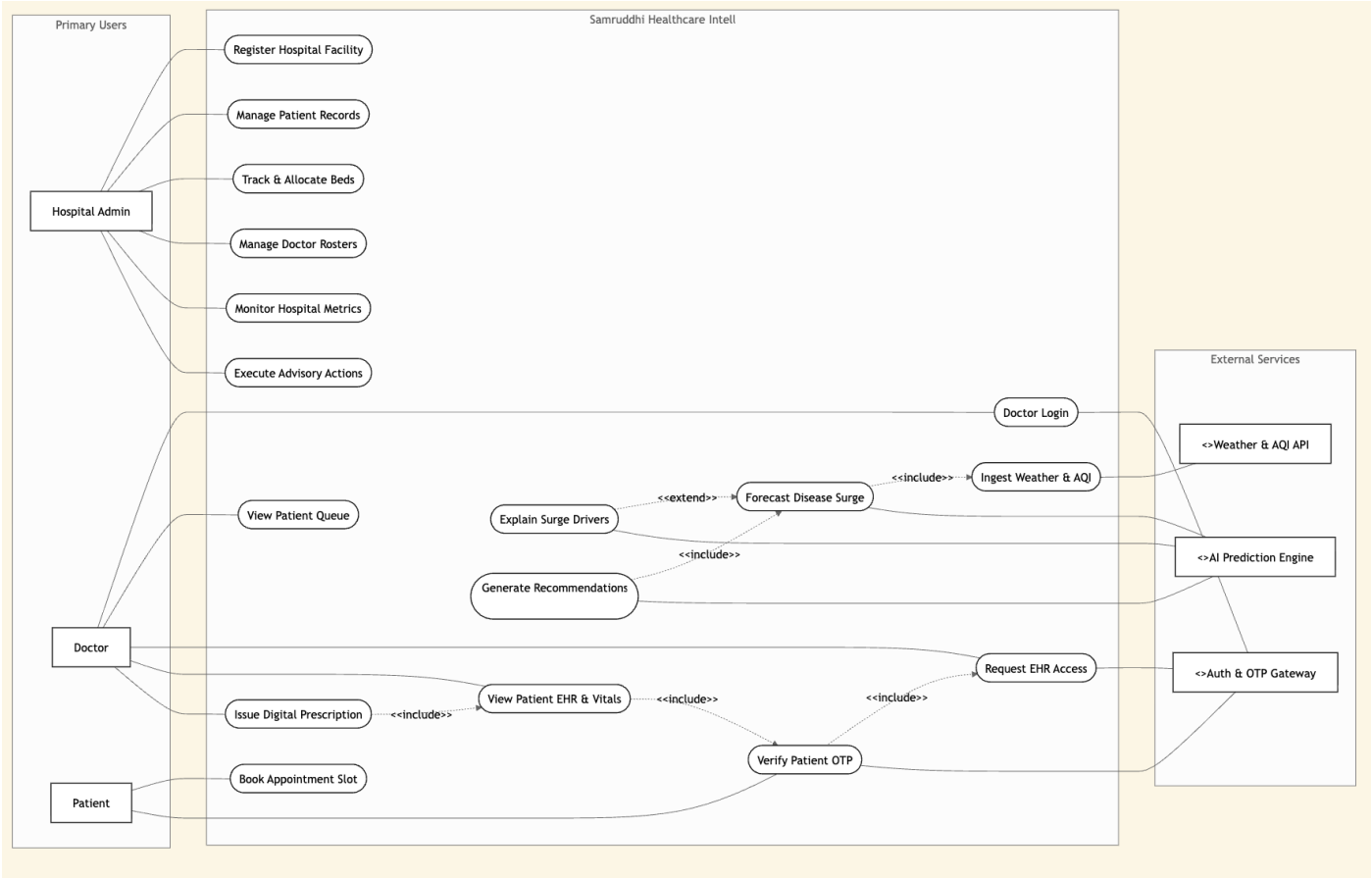


Figure 1: Use Case Diagram for Samruddhi Healthcare Intelligence Platform

Use Case Architectural Highlights & Actor Analysis

- **Hospital Admin (Primary Actor):** Manages facility registrations, tracks and allocates hospital beds, oversees doctor rosters, monitors hospital metrics, and executes AI-generated advisory actions.
- **Doctor (Primary Actor):** Authenticates via Doctor Login, monitors live patient queue, requests patient EHR records, and issues digital prescriptions.
- **Patient (Primary Actor):** Schedules appointments, receives SMS verification tokens, and validates OTP to grant explicit consent for medical record access.
- **External Services:** Integrates with *Weather & AQI API* (environmental feeds), *AI Prediction Engine* (surge forecasting), and *Auth & OTP Gateway* (two-factor authentication).
- **Stereotype Relationships:** *Forecast Disease Surge* <<include>> *Ingest Weather & AQI*; *Explain Surge Drivers* <<extend>> *Forecast Disease Surge*; *View Patient EHR* <<include>> *Verify Patient OTP*.

Diagram 2: Activity Diagram (Operational Workflow)

The Activity diagram illustrates the end-to-end clinical consultation process across three swimlanes: **Patient**, **Doctor & UI Portal**, and **AI Engine & Database**.

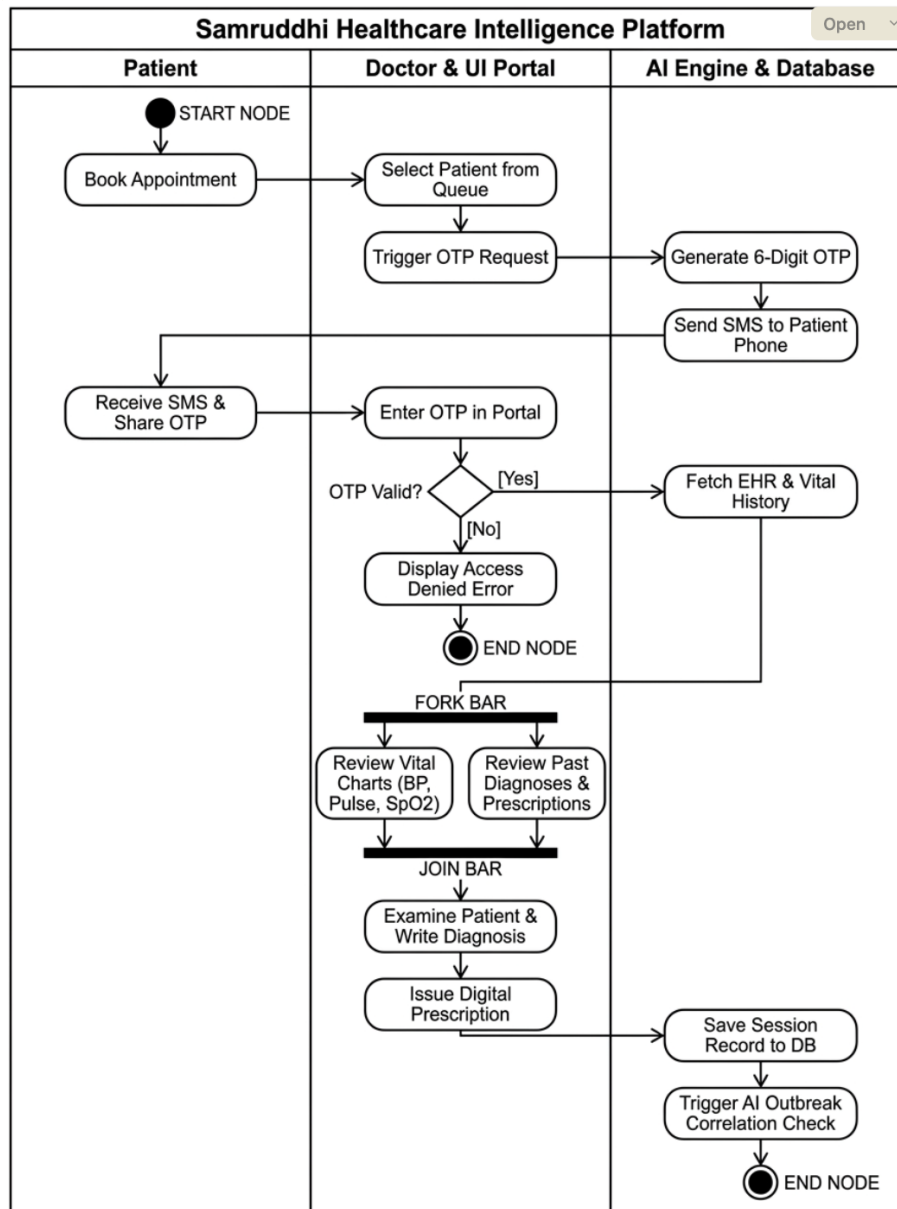


Figure 2: Activity Diagram illustrating Consultation Lifecycle & Outbreak Correlation

Activity Workflow & Control Logic Breakdown

- **Initial Initiation:** Workflow begins at the Patient Start Node upon booking an appointment and joining the clinic queue.
- **Zero-Trust OTP Authentication:** Doctor selects patient → Triggers OTP request → System generates 6-digit OTP and sends SMS to patient phone → Patient shares OTP → Doctor inputs code in UI portal.
- **Decision Gate [OTP Valid?]:** If invalid, an 'Access Denied' error displays and the session reaches an End Node. If valid, EHR & vital history is securely fetched.
- **Concurrent Synchronization (Fork & Join):** Doctor simultaneously reviews vital charts (BP, Pulse, SpO2) and past medical histories/prescriptions before diagnosing.
- **Post-Consultation AI Hook:** Upon issuing the digital prescription, session data is committed to the database, asynchronously triggering the AI Outbreak Correlation check.

Diagram 3: Sequence Diagram (Dynamic Interaction)

The Sequence diagram models the temporal message passing sequence and consent handshake between distributed components during patient consultation.

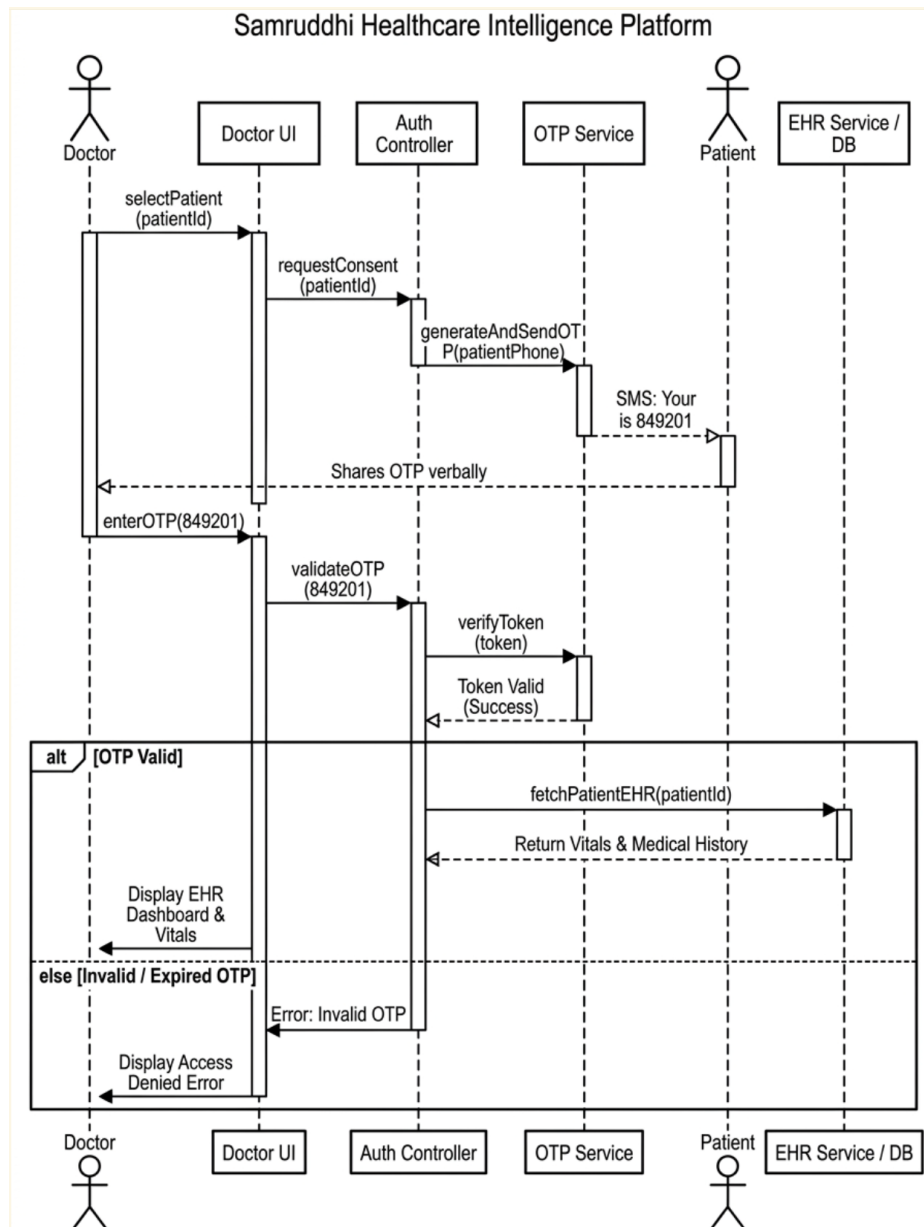


Figure 3: Sequence Diagram for Zero-Trust Patient Consent & EHR Retrieval

Temporal Interaction & Message Protocol Analysis

- **Participating Lifelines:** Doctor (Actor), Doctor UI (Frontend), Auth Controller (API), OTP Service (Auth), Patient (Actor), and EHR Service / DB (Storage).
- **Consent Request Flow:** Doctor invokes `selectPatient(patientId)` → Doctor UI issues `requestConsent(patientId)` to Auth Controller → Controller delegates to OTP Service.
- **Out-of-Band SMS Dispatch:** OTP Service dispatches SMS directly to Patient (e.g., 'SMS: Your OTP is 849201') → Patient verbally communicates OTP to Doctor.
- **Combined Fragment [alt / else]:**
 - **[OTP Valid]:** Auth Controller requests `fetchPatientEHR(patientId)` from EHR Service and returns full vitals and clinical history to Doctor UI.
 - **[Invalid / Expired OTP]:** Auth Controller returns error response, and Doctor UI immediately presents an 'Access Denied Error'.

Diagram 4: Class Diagram (Static Domain Model)

The Class diagram defines the static structural schema, object encapsulation, inheritance, aggregations, compositions, and multiplicity relationships in the domain layer.

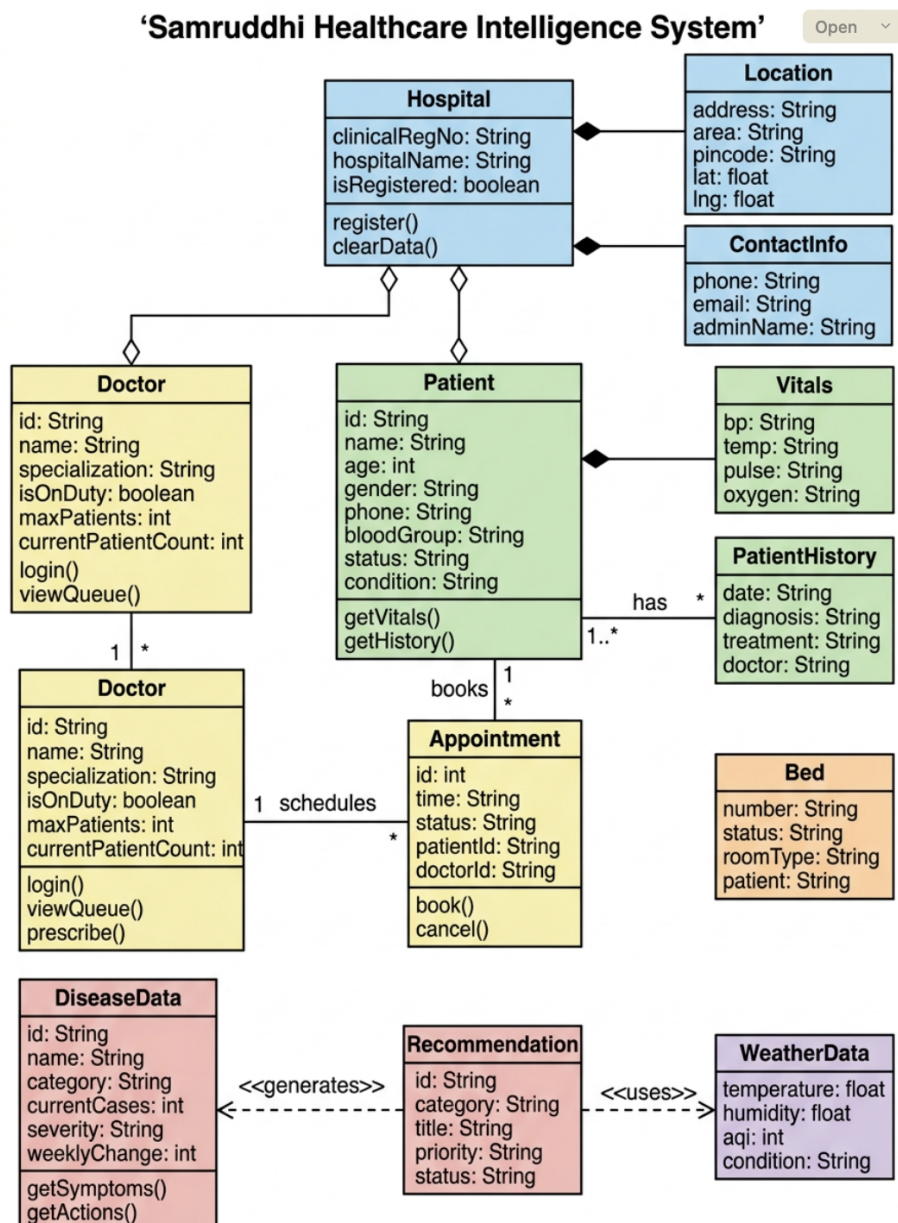


Figure 4: Class Diagram showing Domain Entities, Attributes & Multiplicities

Object Model & Structural Relationship Specifications

- **Hospital Root Aggregate:** Composed of Location (address, geo-coordinates) and ContactInfo (phone, email, adminName); aggregates Doctor and Patient entities.
- **Patient & Clinical Telemetry:** Patient has a strong composition relationship with real-time Vitals (BP, temperature, pulse, oxygen) and a 1..* association with PatientHistory records.
- **Doctor & Scheduling:** Doctor (1) manages Appointment (*), which is booked by Patient (1). Includes methods: login(), viewQueue(), prescribe().
- **Resource Management:** Bed class tracks hospital bed inventory (number, status, roomType, patient).
- **Epidemiological Intelligence:** Recommendation uses WeatherData (temperature, humidity, aqi) to generate dynamic insights for DiseaseData.

Diagram 5: Deployment Diagram (Physical Cloud Architecture)

The Deployment diagram depicts the physical infrastructure, cloud containerization, multi-tier hosting nodes, and network communication security protocols.

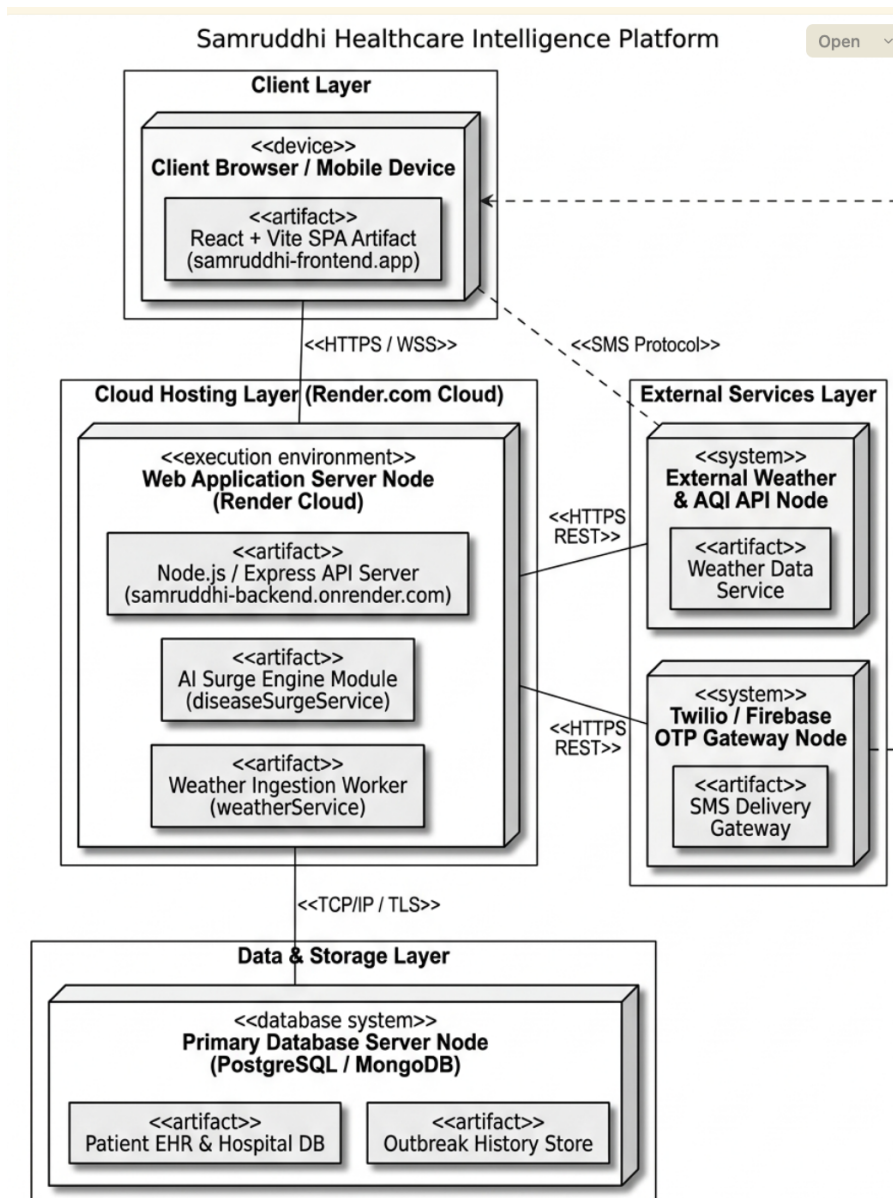


Figure 5: Physical Cloud Deployment Architecture on Render.com Cloud

Physical Infrastructure & Network Protocol Mapping

- **Client Layer:** Client Browser / Mobile Device running the React + Vite Single Page Application (SPA) artifact (samruddhi-frontend.app).
- **Cloud Hosting Layer (Render.com Cloud):** Web Application Server Node running Node.js / Express API Server, housing the AI Surge Engine Module and Weather Ingestion Worker.
- **External Services Layer:** External Weather & AQI API Node (REST) and Twilio / Firebase OTP Gateway Node communicating via SMS protocol to client devices.
- **Data & Storage Layer:** Primary Database Server Node (PostgreSQL / MongoDB) maintaining Patient EHR, Hospital DB, and Outbreak History Store.
- **Network Transport Security:** `<<HTTPS / WSS>>` for client-server traffic, `<<TCP/IP / TLS>>` for encrypted database connectivity, and `<<HTTPS REST>>` for external services.

6. Architectural Synthesis & Traceability Matrix

The following matrix summarizes how the five UML models form a cohesive, robust, and traceable software engineering blueprint for the Samruddhi Healthcare Intelligence Platform:

UML Model	Architectural Viewpoint	Primary Artifacts & Elements	Core Engineering Goal
1. Use Case	Functional & User Boundary	Admin, Doctor, Patient, External APIs	Define scope, actor roles & permissions
2. Activity	Operational & Workflow Logic	Swimlanes, OTP Gate, Fork/Join Vitals	Model business process & async AI hooks
3. Sequence	Temporal Component Interactivity	Doctor UI, Auth Controller, OTP, EHR DB	Enforce zero-trust patient consent protocol
4. Class	Static Object & Data Domain	Hospital, Doctor, Patient, DiseaseData	Establish OOP schema & entity relationships
5. Deployment	Physical Topology & Infrastructure	Render Node.js, React SPA, Mongo/Postgres	Map hardware nodes, cloud hosting & TLS

7. Technology Stack & Runtime Environments

Tier / Layer	Technologies & Frameworks	Architectural Responsibility
Frontend (Client)	React.js, Vite, Modern CSS / Tailwind	Ultra-fast Single Page Application (SPA), responsive dashboard
Backend API Server	Node.js, Express.js REST API Server	Session validation, consultation queuing, and EHR orchestration
Intelligence Layer	Python / Node.js AI Surge Engine	Multi-variate regression and epidemic surge forecasting
Database Layer	PostgreSQL / MongoDB Atlas	ACID transactional patient records and time-series weather store
Cloud & Third-Party	Render.com, Twilio SMS, OpenWeather API	Auto-scaling cloud containerization, OTP delivery, ambient feeds

8. Engineering Conclusion

The **Samruddhi Healthcare Intelligence Platform** architecture provides an enterprise-ready blueprint combining rigorous patient data security with predictive AI healthcare analytics. Through its progressive modeling methodology—from high-level Use Cases to granular Cloud Deployment—the system guarantees end-to-end traceability, high maintainability, and seamless cloud scalability.